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Acyclic Quivers of Finite Mutation Type

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The theory of cluster algebras gives rise to an important notion of quiver mutations. Using the representation theory of finite-dimensional algebras, we show that the acyclic connected quivers of finite mutation type with at least three vertices are precisely the Dynkin and extended Dynkin quivers with arbitrary orientation.

1 Introduction

Fomin and Zelevinsky [13] have defined cluster algebras and developed an interesting and influential theory about this class of algebras. There were originally two major motivations for this theory. The first is to understand the canonical (or global crystal) basis [22, 25, 26] of quantized enveloping algebras associated to a semisimple complex Lie algebra. The second is to describe totally positive elements in a reductive algebraic group [27]. Later, the theory has turned out to be relevant for numerous fields of mathematics.

An essential ingredient of the definition of cluster algebras is an operation on square integral matrices, called matrix mutation. We deal here with the case of skew-symmetric $n \times n$ matrices. Then for each row $k \in \{1, \dots, n\}$ in a matrix $B = (b_{ij})$, there is associated a new matrix $\mu_k(B) = (b'_{ij})$ defined by $b'_{ij} = -b_{ij}$ if $i = k$ or $j = k$, and $b'_{ij} = b_{ij} + (1/2)(|b_{ik}|b_{kj} + b_{ik}|b_{kj}|)$ otherwise. Continuing the process of repeated *mutation in direction* k for $k = 1, \dots, n$, we obtain a collection of integral skew-symmetric $n \times n$-matrices associated with the given matrix B. This collection is called the *mutation class* of B.

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There is a natural one-one correspondence between integral skew-symmetric $n \times n$-matrices and finite quivers with n vertices, having no oriented cycles of length at most two, and hence mutation can be defined directly on the quiver level. If we start with a matrix B corresponding to a Dynkin quiver, it follows from [16] that the mutation class of B is finite.

The purpose of this paper is to use the well-developed representation theory of tame and wild hereditary finite-dimensional algebras, combined with results from recent theory of cluster categories and cluster-tilted algebras [6–8], to prove the following. Given a matrix corresponding to a connected quiver without oriented cycles, the mutation class is finite if and only if the quiver is Dynkin, extended Dynkin, or contains at most two vertices. This was conjectured by A. Seven, who has shown one implication (that finite mutation type implies Dynkin, extended Dynkin, or rank 2) using different methods [33].

In the language of cluster algebras our main result says that acyclic cluster algebras are defined in terms of only a finite number of matrices if and only if at least one of the associated quivers without oriented cycles is Dynkin, extended Dynkin, or has at most two vertices.

The first link from cluster algebras to tilting theory for finite-dimensional algebras was discovered by Marsh et al. [28]. This inspired the invention of cluster categories and cluster-tilted algebras, which is essential for the proof of our theorem. The links from cluster algebras to other fields of mathematics and physics are already numerous. The theory was used in a solution [15] of a conjecture of Zamolodchikov concerning Y-systems, a class of functional relations used in the theory of the thermodynamic Bethe Ansatz, as well as in a solution [14] of various recurrence problems involving Laurent polynomials, including a conjecture of Gale and Robinson on the integrality of generalized Somos sequences. There are links to Poisson geometry [17] and Teichmüller spaces [18], to positive spaces and stacks [12], to dual braid monoids [4] and to ad-nilpotent ideals of a Borel subalgebra of a simple Lie algebra [29]. In addition we note that the matrix mutation formula appears in string theory; see [9].

2 Background

In this section we collect some background material on finite-dimensional hereditary algebras, cluster algebras, and on cluster categories and cluster-tilted algebras.

Let K be an algebraically closed field, and Q a finite connected quiver with no oriented cycles. The associated path algebra KQ is a finite-dimensional hereditary algebra.

For such algebras there is a well-developed module theory; see [1, 31]. In particular there is a nice characterization of finite, tame, and wild representation type in this case.

Theorem 2.1. Let KQ be as above. Then,

(a) KQ is of finite representation type if and only if the underlying graph of Q is Dynkin;

(b) KQ is of tame representation type if and only if the underlying graph is extended Dynkin. □

In the case of infinite type there are three kinds of indecomposable modules: preprojective, preinjective, and regular. The preprojective ones are those of the form $\tau^{-i}P$ for $i \geq 0$, where P is projective and τ is the AR-translation. The preinjective ones are those of the form $\tau^{i}I$ for $i \geq 0$, where I is injective, and the regular ones are the rest. All the preprojective (preinjective) indecomposables are in the same component of the AR-quiver, this is called the preprojective (preinjective) component. For the regular ones, the components of the AR-quiver are tubes in the tame case and of the form $\mathbb{Z}A_\infty$ in the wild case. In each case one has the concept of *quasi-length* for regular modules. Informally, the quasi-length of an indecomposable regular module is a measure of how far the module is from the border of the component. See [24] for details.

A KQ-module X is said to be exceptional if $\mathrm{Ext}^1_{KQ}(X, X) = 0$. There is the following useful information about such modules, where part (b) is from [19].

Proposition 2.2. Let KQ be as above, and X an indecomposable KQ-module.

(a) If X is preprojective or preinjective, then X is exceptional.

(b) If X is regular and exceptional, then X has quasi-length at most $n - 2$, where n denotes the number of vertices in the quiver. □

A KQ-module T is a *tilting module* if T is exceptional and there is an exact sequence $0 \to KQ \to T_0 \to T_1 \to 0$, with T_0 and T_1 direct summands of finite direct sums of copies of T. We have the following [5, 21].

Proposition 2.3. (a) If T is a tilting KQ-module, then the number of nonisomorphic indecomposable direct summands of T is equal to the number of vertices of Q.

(b) Any exceptional KQ-module can be extended to a tilting module. □

We next give some informal background on cluster algebras [13]. The cluster algebras considered here are of the following special type; there are no coefficients and the exchange matrix is skew-symmetric (see [13] for the general set up). Let n be a positive integer and $B = (b_{ij})$ a skew-symmetric integral $n \times n$-matrix. Associated with B is a quiver $Q = Q_B$ with vertices $1, \ldots, n$ corresponding to the rows of the matrix.

Then Q has no loops and no oriented cycles of length two. If $b_{ij} > 0$, there are b_{ij} arrows from i to j. Let $\mathbb{F} = \mathbb{Q}(u_1, \ldots, u_n)$, where the u_i are indeterminates. Start with a so-called seed $(\underline{x}, B)$, where $\underline{x} = \{x_1, x_2, \ldots, x_n\}$ is a free-generating set and B an integral skew-symmetric $n \times n$-matrix. For each index $k = 1, \ldots, n$, we define a new seed $\mu_k(\underline{x}, B) = (\underline{x'}, B')$ as follows. First define a new element $x'_k \in \mathbb{F}$ via the relation: $x'_k x_k = \prod_{x \in \underline{x}, b_{ik} > 0} x^{b_{ik}} + \prod_{x \in \underline{x}, b_{ik} < 0} x^{-b_{ik}}$. Here, we say that x_k and x'_k form an *exchange pair*. Let $\underline{x}' = \underline{x} \cup \{x'_k\} \setminus \{x_k\}$, this is a new free-generating set for $\mathbb{F}$. Let $B' = (b'_{ij})$ be the *mutation* of the matrix B in direction k (as defined in [13]), that is,

$$b'_{ij} = \begin{cases} -b_{ij} & \text{if } i = k \text{ or } j = k, \\ b_{ij} + \frac{1}{2}\left(|b_{ik}|b_{kj} + b_{ik}|b_{kj}|\right) & \text{otherwise.} \end{cases} \tag{2.1}$$

The pair $(\underline{x'}, B')$ is called the *mutation* of the seed $(\underline{x}, B)$ in direction k. The free-generating sets obtained this way, by iterating mutations in all directions, are called *clusters*, and the elements in the clusters are called *cluster variables.* The associated cluster algebra is the subalgebra of $\mathbb{F}$ generated by the cluster variables. The case where one of the matrices occurring in a seed corresponds to a quiver without oriented cycles gives by definition an *acyclic cluster algebra* [2].

Note that the mutation operation is defined on skew-symmetric integral matrices. Such matrices correspond to finite quivers with no loops or oriented cycles of length two. We let the *mutation class* of such a quiver Q denote the set of all quivers which can be reached from Q by a finite number of mutations.

A major result is the following [16], where a cluster algebra is said to be of finite type if there are only a finite number of seeds.

Theorem 2.4. The following are equivalent for a cluster algebra.

(a) There are only a finite number of seeds.

(b) There are only a finite number of clusters.

(c) One of the matrices occurring in a seed is associated with a Dynkin quiver.

□

In [34], Seven gives a list of cluster algebras of minimal infinite type. This can be used to determine if a given cluster algebra is of finite type. If a cluster algebra is of finite type, there is also a finite number of associated nonisomorphic quivers. But the converse is not true, and the aim of this paper is to characterize the class of acyclic cluster algebras which have only a finite number of associated nonisomorphic quivers.

The cluster categories associated with finite-dimensional hereditary algebras were introduced and investigated in [6], trying to model some of the concepts from cluster algebras in a categorical/module theoretical way. Let as before K be an algebraically closed field, Q a finite quiver without oriented cycles, and $H = KQ$ the associated path algebra. Then the associated cluster category is by definition $\mathcal{C}_H = D^b(H)/\tau^{-1}[1]$, where $D^b(H)$ is the bounded derived category of the finitely generated H-modules and $[1]$ the shift functor in $D^b(H)$. Then $\mathcal{C}_H$ is a triangulated category [23]. (Cluster-)tilting objects in $\mathcal{C}_H$ are defined to be basic objects T such that $\mathrm{Ext}^1_{\mathcal{C}_H}(T, T) = 0$, and such that T is maximal with respect to this property. The tilting objects in $\mathcal{C}_H$ are the analogs of clusters, and the indecomposable objects in $\mathcal{C}_H$ are the analogs of cluster variables. The tilting objects in $\mathcal{C}_H$ are the objects induced (via a canonical functor $\mathrm{mod}\, H' \to \mathcal{C}_H$) by a tilting module over some hereditary algebra H' derived equivalent to H. Their endomorphism algebras are the cluster-tilted algebras investigated in [7, 8].

In the next section we use the theory of cluster categories and cluster-tilted algebras, along with results on hereditary algebras, to prove our main result. A crucial role in this investigation is played by the following [7].

Theorem 2.5. Let Q be a finite quiver without oriented cycles. Then the mutation class of Q coincides with the set of quivers of cluster-tilted algebras determined by KQ. □

3 Main result

This section is devoted to proving when a quiver without oriented cycles has a finite mutation class, and thus proving when acyclic cluster algebras are defined using only a finite number of nonisomorphic quivers. This is obtained as a direct consequence of the following result on cluster-tilted algebras.

Theorem 3.1. Let K be an algebraically closed field and Q a connected finite quiver without oriented cycles. Then the following are equivalent for $H = KQ$.

(a) There are only a finite number of (basic) cluster-tilted algebras associated with H, up to isomorphism.

(b) There are only a finite number of quivers, up to isomorphism, occurring for cluster-tilted algebras associated with H.

(c) H is of finite or tame representation type, or has at most two nonisomorphic simple modules. □

Proof. (a) implies (b). This is obvious.

(c) implies (a). If H is of finite representation type, the statement in (a) is obvious, since there are only a finite number of nonisomorphic tilting objects in this case.

For the remaining cases, the following easily seen fact is useful. ■

Lemma 3.2. For any integer n, and any object X in $\mathcal{C}_H$, $\mathrm{End}_{\mathcal{C}_H}(X) \simeq \mathrm{End}_{\mathcal{C}_H}(\tau^n X)$. □

Proof. This follows directly from the fact that $\tau : \mathcal{C}_H \to \mathcal{C}_H$ is an autoequivalence. ■

Assume next that there are at most two (nonisomorphic) simple H-modules. If there is only one simple module, we have $KQ \simeq K$. So we can assume there are two simple H-modules. Then there is no indecomposable regular module which is exceptional, by Proposition 2.2. Up to τ-shift in the cluster category $\mathcal{C}_H$ we can assume that a tilting object in $\mathcal{C}_H$ is induced by a tilting H-module T with an indecomposable projective direct summand P. Then the almost complete tilting object P has exactly two complements in $\mathcal{C}_H$ by [6]. Since there are two possible choices for P, we obtain that there are at most four quivers of cluster-tilted algebras.

Assume now that H is tame. We show that there is a finite set $\mathcal{M} = \{M_i\}_{i \in I}$, such that for any tilting object T in $\mathcal{C}_H$, there is an integer j and an $i \in I$, such that $T \simeq \tau^j M_i$. We need the following.

Lemma 3.3. For each indecomposable projective H-module P, the set of objects X in $\mathcal{C}_H$ with $\mathrm{Ext}^1_{\mathcal{C}}(X, P) = 0$ is finite. □

Proof. All indecomposable objects in $\mathcal{C}_H$ are either isomorphic to H-modules (via the canonical map $\mathrm{mod}\, H \to \mathcal{C}_H$) or isomorphic to objects of the form $P'[1]$, where P' is indecomposable projective. It is therefore clearly sufficient to show that there are only a finite number of H-modules X with $\mathrm{Ext}^1_{\mathcal{C}}(X, P) = 0$. We have $\mathrm{Ext}^1_H(X, P) \simeq D\,\mathrm{Hom}_H(P, \tau X)$, where D denotes the duality for $\mathrm{mod}\, H$. Let S be the simple top of P. Then we have that $\mathrm{Hom}_H(P, Y) = 0$ if and only if S is not a composition factor of Y. For tame hereditary algebras, there are only a finite number of indecomposable modules which do not have S as a composition factor. Thus, the claim follows. ■

It follows from Lemma 3.3 that there are only a finite number of tilting objects with an indecomposable projective H-module P as a direct summand, and thus the set $\mathcal{M}$ of tilting objects in $\mathcal{C}_H$ with a projective H-module as a direct summand is finite. It is known [20] that T has at least one direct summand which is preprojective or preinjective. This finishes the proof that (c) implies (a).

For proving that (b) implies (c), the following two lemmas are useful.

Lemma 3.4. Assume H is wild with at least three nonisomorphic simple H-modules. Given $m > 0$, there is a tilting object T in $\mathcal{C}_H$ with indecomposable direct summands T_1, T_2 such that $\dim \mathrm{Hom}_{\mathcal{C}_H}(T_1, T_2) \geq m$. □

Proof. Given the assumptions on H, there is a vertex e in the quiver of H such that $A = H/HeH$ is of infinite representation type. By possibly replacing H by a derived equivalent hereditary algebra, we can assume that e is a source in the quiver. This can be seen as follows. Consider the set $\mathcal{P}$ of paths ρ in Q (where $H = KQ$) with the following properties:

(i) the path ρ ends in e;

(ii) the path ρ is not a proper subpath of a path ρ' ending in e.

Let m_H be the sum of the lengths of the paths in $\mathcal{P}$. Assume $m_H > 0$. Then there is at least one path in $\mathcal{P}$, assume this path starts in f. Then f is a source in the quiver Q. Consider the quiver Q' obtained by reversing all arrows starting in f (this is a Bernšteĭn-Gel′fand-Ponomarev reflection [3]). Then $KQ' = H'$ is derived equivalent to H, and $m_{H'} < m_H$. Repeating this a finite number of times (at most m_H), we obtain an algebra H'' derived equivalent to H, and with $m_{H''} = 0$. That is, e is a source for this algebra.

Choose A' to be a connected summand of A of infinite representation type, that is, the quiver of the subalgebra A' is connected, and $A = A' \times A''$ as algebras, for some subalgebra A''. Let S be the simple H-module corresponding to e, and P its projective cover. Choose a vertex e' in the quiver of A' which is a neighbor of e in the quiver of H. Denote by P' the corresponding indecomposable projective A'-module and let S' be the simple top of P'. Let $X_i = \tau_{A'}^{-i} P'$ for $i \geq 0$. Then $\mathrm{Ext}^1_{A'}(X_i, X_i) = 0$ and hence $\mathrm{Ext}^1_H(X_i, X_i) = 0$. Furthermore, $\mathrm{Ext}^1_H(P, \tau_H^{-1} X_i) = 0$, and $\mathrm{Ext}^1_H(\tau_H^{-1} X_i, P) \simeq D\,\mathrm{Hom}_H(P, X_i) = 0$, since S is not a composition factor of the A'-module X_i. Hence $P \oplus \tau_H^{-1} X_i$ is an exceptional H-module for all $i \geq 0$. For each $i \geq 0$ choose a tilting H-module Y_i extending $P \oplus \tau_H^{-1} X_i$.

Given $m > 0$, there is some i_0 such that the simple A'-module S' has multiplicity at least m in X_{i_0} (see [10, Theorem 3.5], [11, pages 11-12], [32]). Let C denote the Coxeter transformation, defined on $K_0(\mathrm{mod}\, H)$ (see [3, 30]). Then $C^{-1}[X_{i_0}] = [\tau_H^{-1}(X_{i_0})]$, where $[X]$ denotes the image of the H-module X in the Grothendieck group $K_0(\mathrm{mod}\, H)$. In particular, the multiplicity of S' in $\tau_H^{-1}(X_{i_0})$ is the coefficient of $[S']$ when $C^{-1}[X_{i_0}]$ is expressed in terms of the basis coming from the simple H-modules. Since e is a source in the quiver of H, we can compute this coefficient as follows, by using the definition of the Coxeter transformation as a composition of reflections. Let $e_1, \ldots, e_t$ be the neighbors of e in the quiver of H, with r_i arrows from e to e_i, and let m_i be the multiplicity of the corresponding simple module S_i in X_{i_0}. Then $\sum_{i=1}^t r_i m_i$ gives the multiplicity of S in $\tau_H^{-1}(X_{i_0})$, and this is at least m, since S' is one of the S_i. Hence $\dim \mathrm{Hom}_H(P, \tau_H^{-1} X_{i_0}) \geq m$, and the claim follows since $P \oplus \tau_H^{-1} X_{i_0}$ is a direct summand of a tilting module, and $\dim \mathrm{Hom}_{C_H}(P, \tau_H^{-1} X_{i_0}) \geq \dim \mathrm{Hom}_H(P, \tau_H^{-1} X_{i_0})$. ■

Lemma 3.5. If a path in the quiver of a cluster-tilted algebra goes through two oriented cycles, then it is zero. □

Proof. Let T be a basic tilting module for a hereditary algebra H, with corresponding cluster category $\mathcal{C}_H$, and assume $T = \oplus_{i=1}^{n} T_i$ is the direct sum of indecomposable modules T_i. There are no oriented cycles in a tilted algebra. Thus, we know that one of the maps in an oriented cycle for the cluster-tilted algebra is a map which lifts to a map of the form $T_{i_1} \to \tau^{-1} T_{i_2}[1]$ in $D^b(H)$. Using that any map $T_a \to T_b[2]$ is zero in $D^b(H)$ and thus is zero also in $\mathcal{C}_H$, the claim is proved.

(b) implies (c). Assume that there are only a finite number of quivers (up to isomorphism) occurring for cluster-tilted algebras associated with H. Let n be the number of vertices, and let u be the maximum number of arrows between two vertices in this finite set of quivers. Using Lemma 3.5 we can then conclude that $\dim \operatorname{Hom}_{\mathcal{C}_H}(T_1, T_2) < u^{2n}$ for any two indecomposable direct summands T_1 and T_2 of a tilting object T. Hence, the claim follows using Lemma 3.4, with $m = u^{2n}$. ■

This has the following consequence for cluster algebras.

Theorem 3.6. Let $\mathcal{A}$ be an acyclic cluster algebra defined by a skew-symmetric integral matrix, or equivalently by a quiver Q. If Q is connected, then the following are equivalent.

(a) The mutation class of Q is finite.

(b) Q is either Dynkin, extended Dynkin, or has at most two vertices. □

Proof. This is a direct consequence of Theorems 2.5 and 3.1. ■

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